

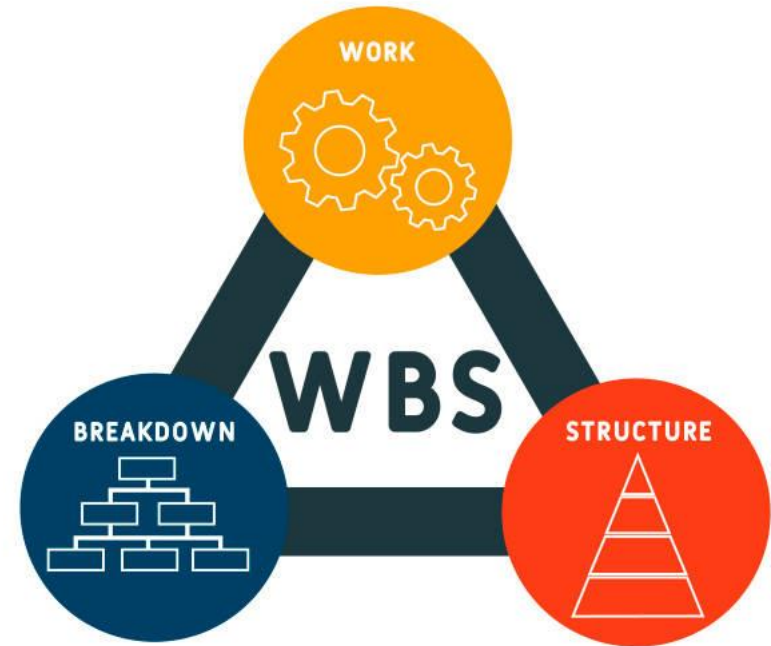
CSE 4101: Project Management

Topic: Project Estimation with WBS Method

Today's Agenda

01

Work Breakdown Structure (WBS)

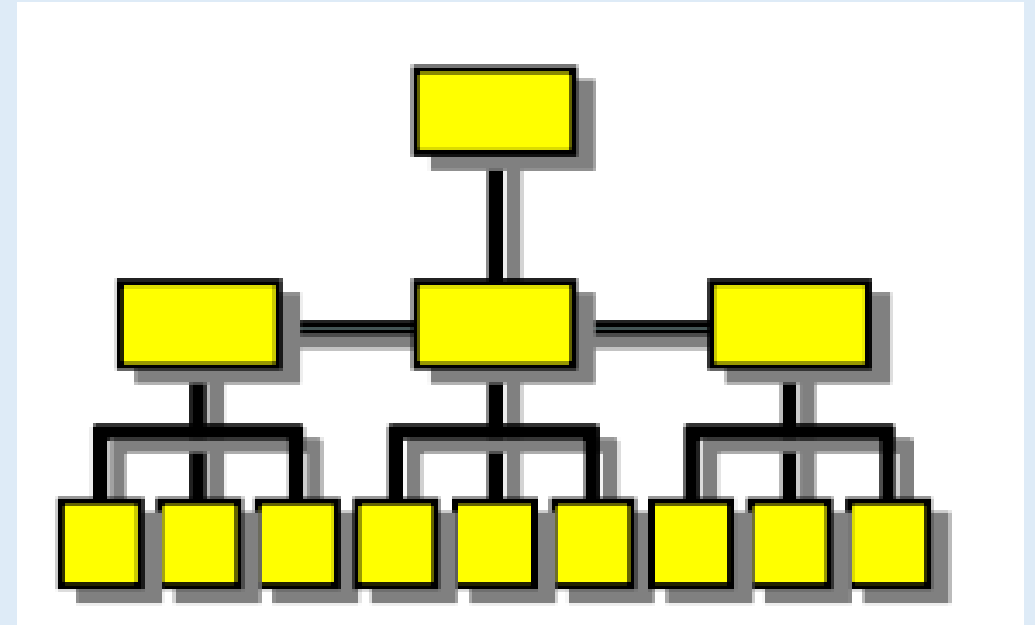


What is Work Breakdown Structure (WBS)?

A work breakdown structure (WBS) is an **outcome-oriented analysis** of the work involved in a project that defines the **total scope** of the project.

What is Work Breakdown Structure (WBS)?

- ✓ Work Breakdown Structure (WBS) provides a notation for representing task structure:
 - ✓ Activities are represented as nodes of a tree.
 - ✓ The root of the tree is labelled by main task
 - ✓ Each task can be broken down into subtasks



Work Breakdown Structure Design Principles

1. Hierarchy

Each Work Breakdown Structure is hierarchical. That means every “child” on the graph has a hierarchical relationship with its parent task. When you add up all the “child” elements, it’ll give you a clear picture of the parent task.

2. 100% rule

While every Work Breakdown Structure is a little different, they all follow the 100% rule. Every level of the graph must make up 100% of the parent level, and it must have at least two “child” elements.

3. Mutually exclusive elements

Every element at each level of a WBS template has to be mutually exclusive. That means there can’t be any overlap between deliverables or work. Enforcing mutual exclusivity helps cut down on miscommunication and avoid duplicate work.

4. Outcome-oriented

WBS is fundamentally a deliverable-oriented system. That means your graphic depiction must focus on the outcomes rather than the activities required to produce them. A good rule of thumb is to describe elements using nouns rather than verbs.

6 tips for making a work breakdown structure

1. The 100% rule

The work represented by your WBS must include 100% of the work necessary to complete the overarching goal without including any extraneous or unrelated work. Also, child tasks on any level must account for all of the work necessary to complete the parent task.

2. Mutually exclusive

Do not include a sub-task twice or account for any amount of work twice. Doing so would violate the 100% rule and will result in miscalculations as you try to determine the resources necessary to complete a project.

6 tips for making a work breakdown structure

3. Outcomes, not actions

Remember to focus on deliverables and outcomes rather than actions. For example, if you were building a bike, a deliverable might be “the braking system” while actions would include “calibrate the brake pads.”

4. The 8/80 rule

There are several ways to decide when a work package is small enough without being too small. This rule is one of the most common suggestions—a work package should take no less than eight hours of effort, but no more than 80. Other rules suggest no more than ten days (which is the same as 80 hours if you work full time) or no more than a standard reporting period. In other words, if you report on your work every month, a work package should take no more than a month to complete.

6 tips for making a work breakdown structure

5. Three levels, Level 2 most important

Generally speaking, a WBS should include about three levels of detail. Some branches of the WBS will be more subdivided than others, but if most branches have about three levels, the scope of your project and the level of detail in your WBS are about right.

6. Make assignments

Every work package should be assigned to a specific team or individual. If you have made your WBS well, there will be no work overlap so responsibilities will be clear.

Other than Deliverables Oriented Other WBSs:

- ✓ **Verb-oriented WBS:** A task-oriented WBS defines the deliverable of project work in terms of the actions that must be done to produce the deliverable. The **first word** in a given WBS element usually is a verb, such as, design, develop, optimize, transfer, test, etc.
- ✓ **Noun-oriented WBS:** a deliverable-oriented WBS defines project work in terms of the components (physical or functional) that make up the deliverable.
- ✓ **Time-phased WBS:** a “time-phased” WBS is one that is used on very long projects. It breaks the project into major phases instead of tasks.

Approaches to Developing WBSs

- ✓ Using guidelines: Some organizations, like the DOD, provide guidelines for preparing WBSs
- ✓ The analogy approach: It often helps to review WBSs of similar projects
- ✓ The top-down approach: Start with the largest items of the project and keep breaking them down
- ✓ The bottoms-up approach: Start with the detailed tasks and roll them up

Bottom-up versus Top-down

Bottom-up

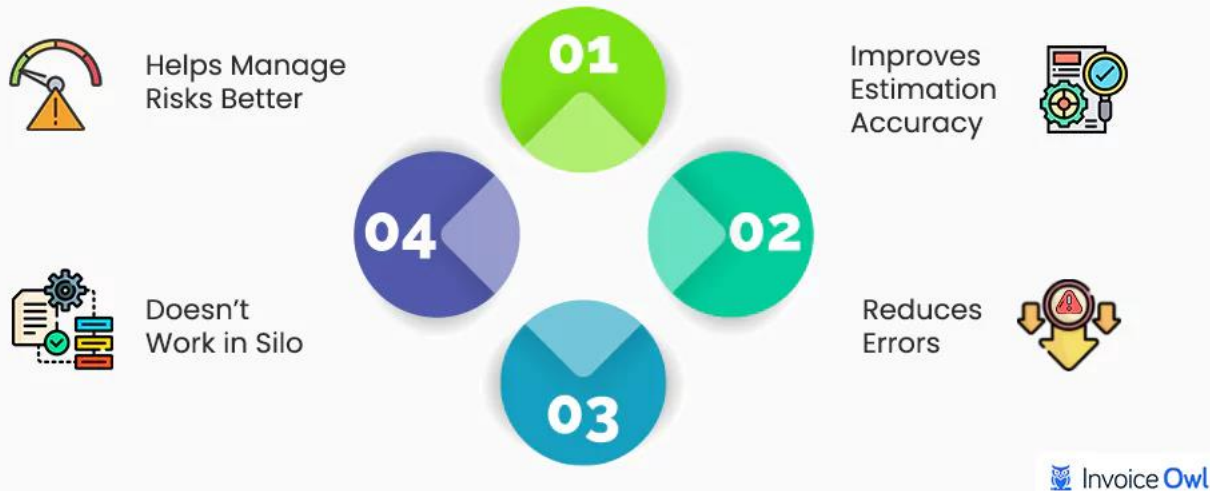
- ✓ use when no past project data
- ✓ identify all tasks that have to be done – so quite time-consuming
- ✓ use when you have no data about similar past projects

Top-down

- ✓ produce overall estimate based on project cost drivers
- ✓ based on past project data
- ✓ divide overall estimate between jobs to be done

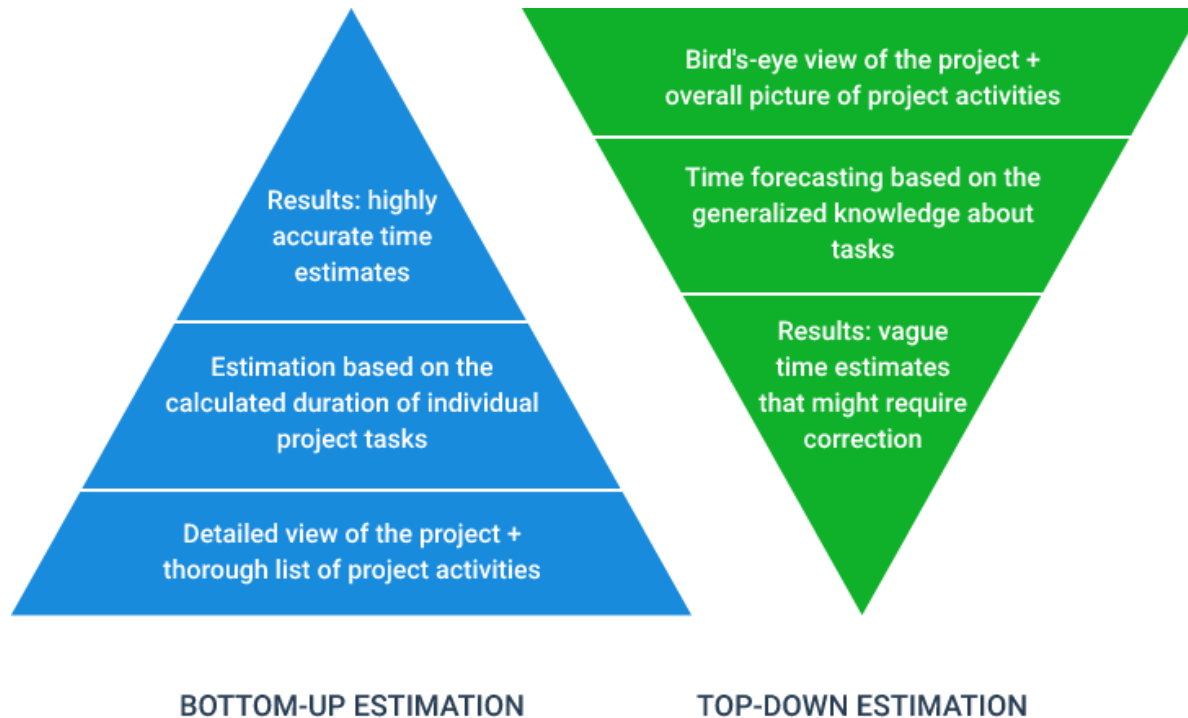
Bottom-up Estimating

Why use Bottom-up Estimating Technique?



1. Break project into smaller and smaller components
2. Stop when you get to what one person can do in one/two weeks
3. Estimate costs for the lowest level activities
4. At each higher level calculate estimate by adding estimates for lower levels

Top-down Estimating



- ✓ Produce overall estimate using effort driver (s)
- ✓ distribute proportions of overall estimate to components

Pros and Cons of WBS

Advantages

- ✓ More likely to obtain a task catalogue that is complete and is composed of non-overlapping tasks
- ✓ WBS represents a structure that can be refined as the project proceeds
- ✓ The structure already suggests the dependencies among the activities

Disadvantages

- Very likely to miss some activities if an unstructured activity list is used

Work breakdown structure formats

You can choose from several different format options when creating a work breakdown structure. The example above uses a tree format, which is the most visual option. It structures the WBS like an org chart and shows the hierarchy of tasks in addition to providing space for additional information about each work package.

WBS - Outline Structure

Build a House

1 Foundation

1.1 Excavate

1.1.1 Dig

1.1.2 Level

1.2 Frae

1.3 Concrete

1.3.1 Pour

1.3.2 Cure

2 Exterior

3 Interior

A text outline is the simplest WBS format. It is easy to put together and shows the hierarchy of tasks. However, it is difficult to add additional information about budget, duration, and assignment using this format.

WBS - Level Structure

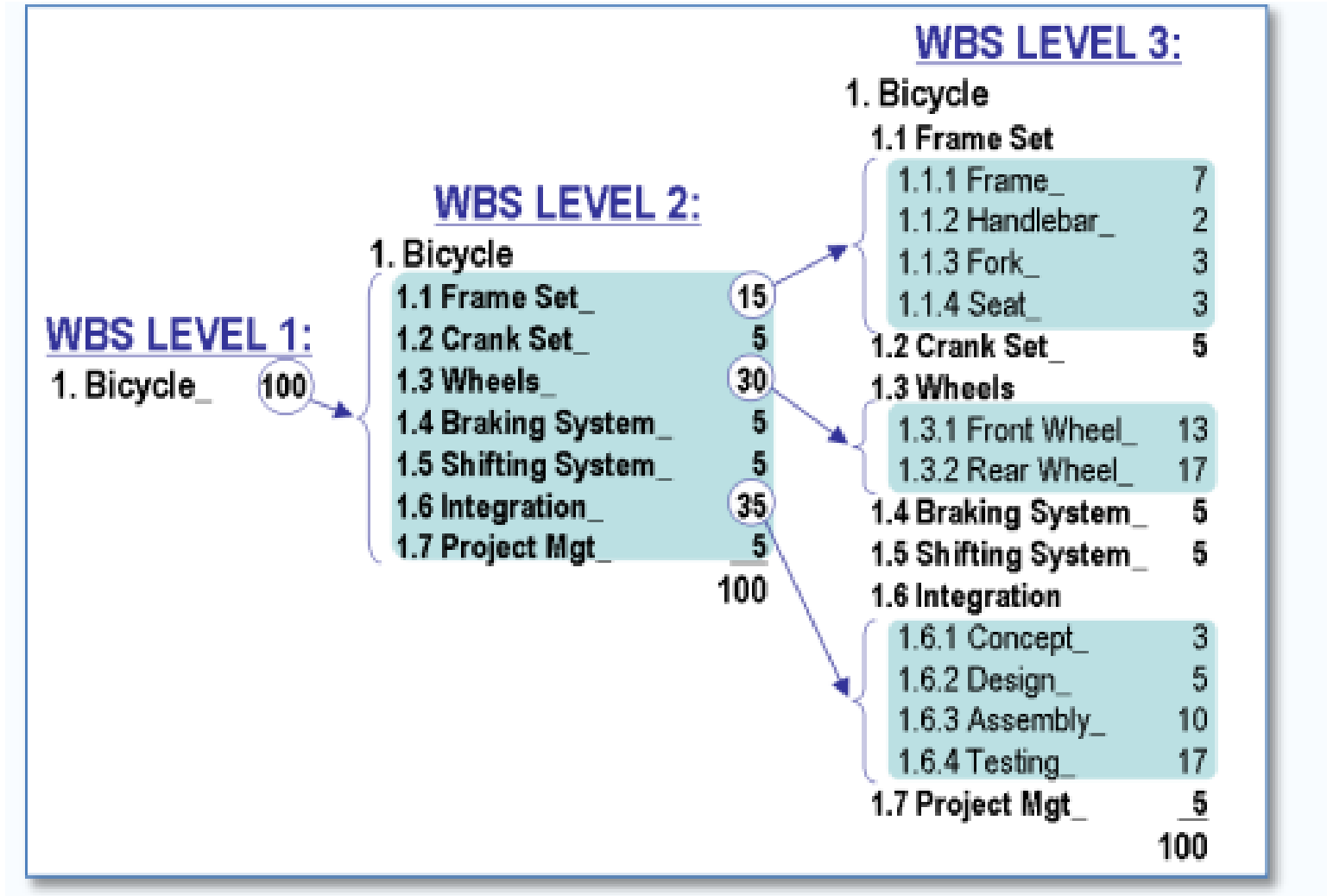


Figure 1 WBS Construction Example

WBS - Tabular Structure

Level 1	Level 2	Level 3
1 Foundation	1.1 Excavate	1.1.1 Dig 1.1.2 Level
	1.2 Frame	
	1.3 Concrete	1.3.1 Pour 1.3.2 Cure
2 Exterior		
3 Interior		

A tabular view is a more visually intuitive way to show hierarchy using a table.

WBS - Dictionary Structure

Level	WBS Code	Element Name	Element Name
1	1	Foundations	All of the work necessary to build a foundation
2	1.1	Excavate	Create a hole ready for the foundation to be framed and poured
3	1.1.1	Dig	Dig a hole of the right shape and size in the correct location
3	1.1.2	Level	Level the hole so that is packed, even, and ready to receive the foundation
2	1.2	Frame	Frame the foundation including steel supports
2	1.3	Concrete	Acquire, transport, pour, and cure the concrete foundation
3	1.3.1	Pour	Pour, pack and level the foundation
3	1.3.2	Cure	All procedures necessary for the foundation to cure successfully
1	2	Exterior	All of the work necessary to complete the exterior of the house
1	3	Interior	All of the work necessary to complete the interior of the house

What is a WBS dictionary? A WBS dictionary is formatted like the hierarchical structure, but it includes a brief description of each work package. When documenting a project, a WBS dictionary is often included in addition to a visualization of the WBS. It helps to clarify the scope of each task so that all team members understand their responsibilities.

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Other Topics:

WBS Pitfalls

WBS Checklist

Thank You